

Snowflake Assessment

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Purpose: This document contains the commands required to create and load the Employee table in Snowflake, copy Employee data to S3, unload the listing table from Redshift to S3 as parquet, and load parquet data into the Snowflake listing table as individual columns.

S3 staging path used: s3://matrika-bucket/Snowflake/

1. Create Employee table in Snowflake and load data

1.1 Create database, schema, and Employee table

```
CREATE DATABASE IF NOT EXISTS bootcamp;

CREATE SCHEMA IF NOT EXISTS bootcamp.snow;

USE DATABASE bootcamp;
USE SCHEMA snow;

CREATE OR REPLACE TABLE Employee (
    EMPLOYEE_ID    INTEGER,
    NAME           VARCHAR,
    SALARY         INTEGER,
    DEPARTMENT_ID  INTEGER,
    JOINING_DATE   DATE
);
```

1.2 Load Employee.csv data into Employee table

```
INSERT INTO Employee
    (EMPLOYEE_ID, NAME, SALARY, DEPARTMENT_ID, JOINING_DATE)
VALUES
    (100, 'Jennifer', 4400, 10, '2017-01-05'),
    (101, 'Michael', 13000, 10, '2018-08-24'),
    (102, 'Pat', 6000, 10, '2018-12-10'),
    (103, 'Den', 11000, 20, '2019-02-17'),
    (104, 'Alexander', 3100, 20, '2019-07-01'),
    (105, 'Shelli', 2900, 20, '2020-04-22'),
    (106, 'Sigal', 2800, 30, '2020-09-05'),
    (107, 'Guy', 2600, 30, '2021-05-25'),
    (108, 'Karen', 2500, 30, '2021-12-21');
```

1.3 Verify Employee table data

```
SELECT *
FROM Employee
ORDER BY EMPLOYEE_ID;
```

Screenshot 1: Employee table data

The screenshot shows the Snowflake Database Explorer interface. On the left, the 'Database Explorer' pane shows a tree view of the database structure, including 'BOOTCAMP', 'INFORMATION_SCHEMA', 'PUBLIC', 'SNOW', 'Tables' (containing 'EMPLOYEE', 'LISTING'), 'Stages' (containing 'EMPLOYEE_OUTPUT_ST...'), 'TEST', 'SNOWFLAKE', 'SNOWFLAKE_SAMPLE_DATA', and 'USER\$MATRIKASUBEDI'. The main pane displays a SQL query: `SELECT * FROM Employee ORDER BY EMPLOYEE_ID;`. Below the query, the 'Results' pane shows a table with 9 rows and 6 columns: `EMPLOYEE_ID`, `NAME`, `SALARY`, `DEPARTMENT_ID`, and `JOINING_DATE`. The results are as follows:

#	EMPLOYEE_ID	NAME	SALARY	DEPARTMENT_ID	JOINING_DATE
1	100	Jennifer	4400	10	2017-01-05
2	101	Michael	13000	10	2018-08-24
3	102	Pat	6000	10	2018-12-10
4	103	Den	11000	20	2019-02-17
5	104	Alexander	3100	20	2019-07-01
6	105	Shelli	2900	20	2020-04-22
7	106	Sigal	2800	30	2020-09-05

2. Copy Employee data from Snowflake to S3 output directory

2.1 Create or replace Snowflake external stage

```
CREATE OR REPLACE STAGE employee_output_stage
URL = 's3://matrika-bucket/Snowflake/'
CREDENTIALS = (
  AWS_KEY_ID = '<your_access_key>',
  AWS_SECRET_KEY = '<your_secret_key>'
);
```

2.2 Unload Employee table data to S3 as CSV

```
COPY INTO @employee_output_stage/employee_data.csv
FROM (
  SELECT
    EMPLOYEE_ID,
    NAME,
    SALARY,
    DEPARTMENT_ID,
    JOINING_DATE
  FROM Employee
)
FILE_FORMAT = (
  TYPE = CSV
  FIELD_DELIMITER = ','
  COMPRESSION = NONE
)
HEADER = TRUE
OVERWRITE = TRUE
SINGLE = TRUE;
```

2.3 Verify file in Snowflake stage and S3

```
LIST @employee_output_stage;
```

Screenshot 2: LIST @employee_output_stage result

Database Explorer

Objects | Data Products

Search

Filter

BOOTCAMP

INFORMATION_SCHEMA

PUBLIC

SNOW

Tables 2

EMPLOYEE

LISTING

Stages 1

EMPLOYEE_OUTPUT_ST...

TEST

SNOWFLAKE

SNOWFLAKE_SAMPLE_DATA

USER\$MATRIKASUBEDI

My Workspace > snowflakeassignment.sql

ACCOUNTADMIN • COMPUTE_WH (X-Small) BOOTCAMP SNOW Share ...

42 HEADER = TRUE

43 OVERWRITE = TRUE

44 SINGLE = TRUE;

45

46 LIST @employee_output_stage;

47

48

Results (1 minute ago)

Table Chart Pivot

27 rows 164ms

	name	size	md5	last_modified
	s3://matrika-bucket/Snowfl...	3.7%	cf2407183d3a9973d80a1cc...	11.1%
	s3://matrika-bucket/Snowfl...	3.7%	333ef743778e35667f22d53...	7.4%
	+25 more		+13 more	
6	s3://matrika-bucket/Snowflake/device...	201	456d06c99d17ff9d07f343aa66c24199	Fri, 29 May 2
7	s3://matrika-bucket/Snowflake/employee...	318	adfd28f2080a091e2f41f5614ee1648b	Fri, 29 May 2
8	s3://matrika-bucket/Snowflake/input/lis	2158	af9045801b9fcdcf8df5c967b95d947e-1	Fri, 29 May 2
9	s3://matrika-bucket/Snowflake/input/lis	2158	e6ffad9d16e6977de54352f321bad079-	Fri, 29 May 2

Feedback

2:51 PM 5/29/2026

Screenshot 3: S3 Employee output file

Amazon S3 > Buckets > matrika-bucket > Snowflake/

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

File systems New

Access management and security

Access Points

Access Points for FSx

Access Grants

IAM Access Analyzer

Objects (1/9)

Create folder Upload

Copy S3 URI Copy URL Download Open Delete Actions

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

	Name	Type	Last modified	Size	Storage class
	device_log.csv	CSV	21:54:02 (UTC-04:00)	201.0 B	Standard
☒	employee_data.csv	csv	May 29, 2026, 14:08:00 (UTC-04:00)	318.0 B	Standard
☐	input/	Folder	-	-	-
☐	listing_parquet/	Folder	-	-	-
☐	patient_vitals.csv	csv	May 27, 2026, 21:34:50 (UTC-04:00)	339.0 B	Standard

CloudShell Feedback Console Mobile App

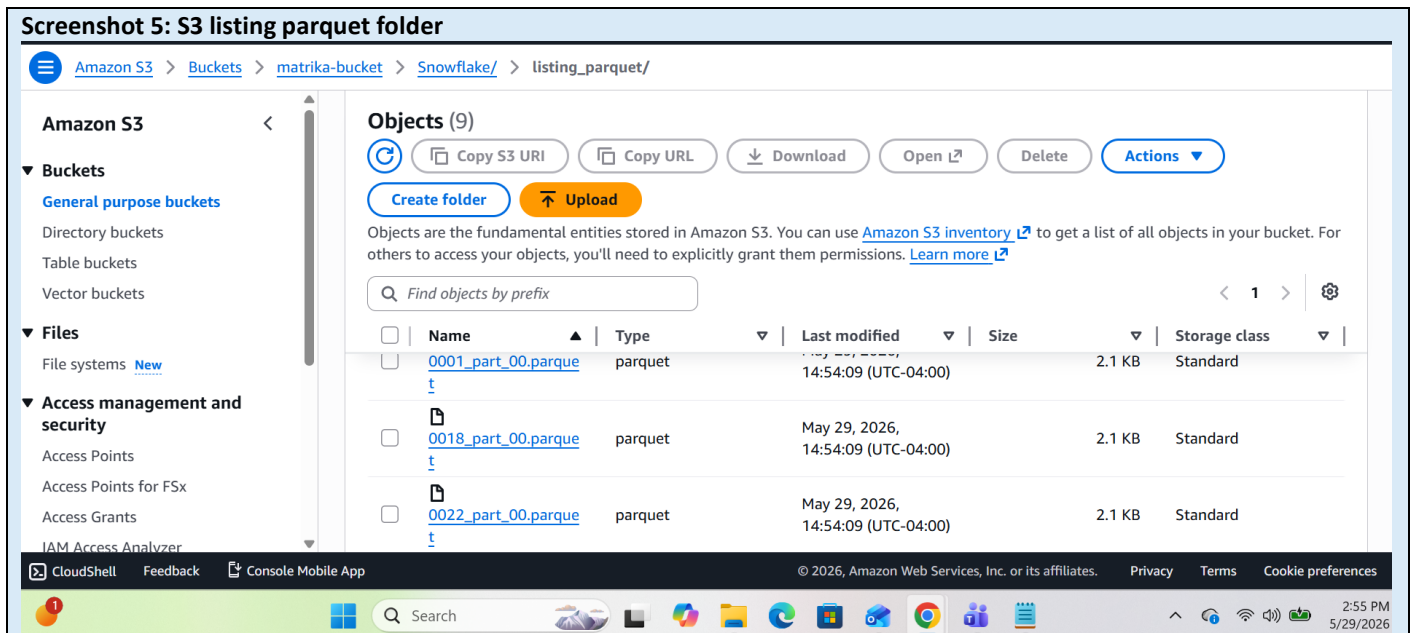
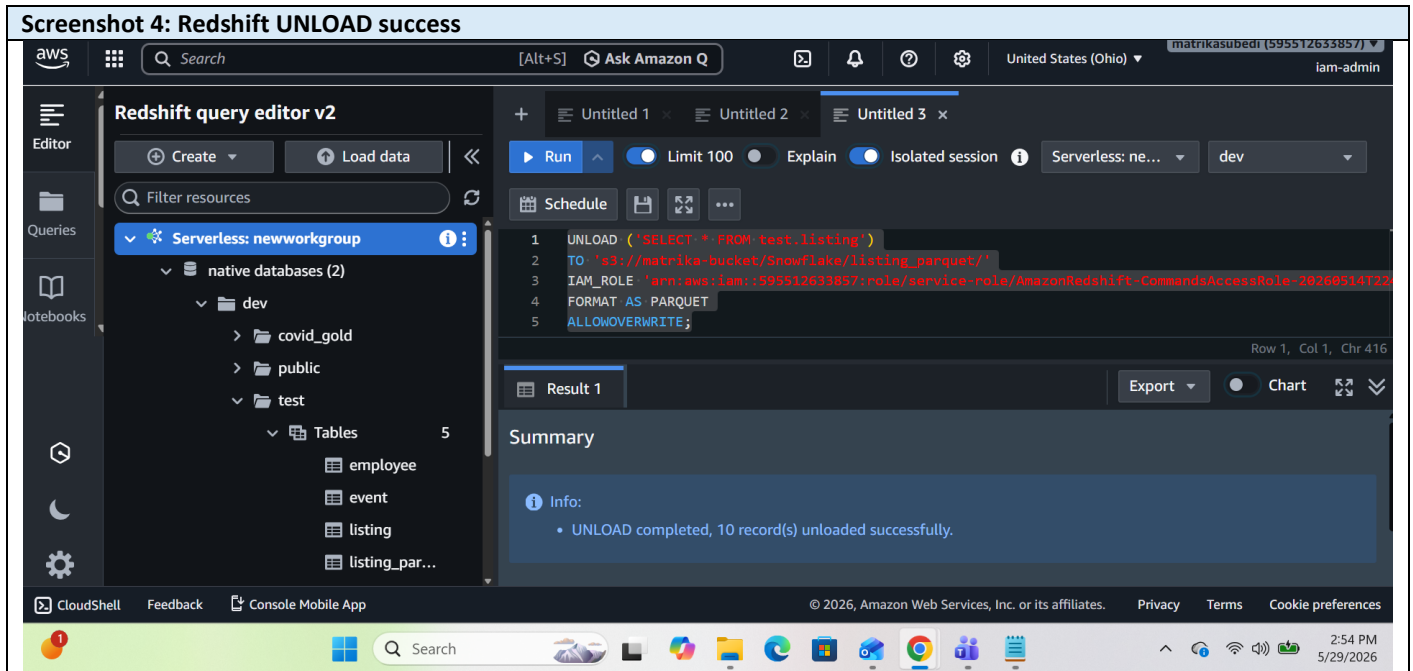
© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

2:52 PM 5/29/2026

3. Redshift unload script for listing table

Run this command in Amazon Redshift. It unloads the listing table into a separate folder inside the Snowflake staging path as parquet files.

```
UNLOAD ('SELECT * FROM test.listing')
TO 's3://matrika-bucket/Snowflake/listing_parquet/'
IAM_ROLE 'arn:aws:iam::595512633857:role/service-role/AmazonRedshift-CommandsAccessRole-20260514T224310'
FORMAT AS PARQUET
ALLOWOVERWRITE;
```



4. Create listing table in Snowflake

The listing table columns are based on the Redshift listing table structure provided.

```
USE DATABASE bootcamp;
USE SCHEMA snow;
```

```
CREATE OR REPLACE TABLE listing (
  listid          INTEGER,
  sellerid        INTEGER,
  eventid         INTEGER,
  dateid          SMALLINT,
  numtickets      SMALLINT,
  priceperticket  NUMBER,
```

```

totalprice      NUMBER,
listtime        TIMESTAMP_NTZ
);

```

5. Load parquet data from S3 into Snowflake listing table

This command uses the same Snowflake stage but points to the listing_parquet/ folder. MATCH_BY_COLUMN_NAME loads values into individual columns, not as JSON records.

```

COPY INTO listing
FROM @employee_output_stage/listing_parquet/
FILE_FORMAT = (
  TYPE = PARQUET
)
MATCH_BY_COLUMN_NAME = CASE_INSENSITIVE;

```

5.1 Verify listing table data

```

SELECT *
FROM listing
LIMIT 10;

```

```

SELECT COUNT(*) AS listing_row_count
FROM listing;

```

Screenshot 6: Listing table sample data

The screenshot shows the Snowflake Data Cloud interface. On the left, the 'Database Explorer' sidebar is open, showing the 'SNOW' database and the 'LISTING' table. The main area displays a SQL query: `SELECT * FROM Listing LIMIT 10;`. Below the query, the results are shown in a table format. The table has 8 columns: LISTID, SELLERID, EVENTID, DATEID, NUMTICKETS, PRICEPERTICKET, TOTALPRICE, and LISTTIME. The results show 10 rows of data.

	LISTID	SELLERID	EVENTID	DATEID	NUMTICKETS	PRICEPERTICKET	TOTALPRICE	LISTTIME
1	5	1616	8647	1963	4	175	700	2008-05-17
2	10	24858	3375	1994	16	197	3152	2008-06-15
3	1	36861	7872	1850	10	182	1820	2008-01-24
4	6	47402	8240	2053	18	77	1386	2008-08-15

Screenshot 7: Listing table row count

The screenshot displays the Snowflake SQL Editor interface. On the left, the 'Database Explorer' sidebar shows a tree view of the database structure, including schemas like 'BOOTCAMP', 'INFORMATION_SCHEMA', 'PUBLIC', and 'SNOW'. The 'SNOW' schema is expanded, showing tables 'EMPLOYEE' and 'LISTING', and stages 'EMPLOYEE_OUTPUT_ST...' and 'TEST'. The main editor area shows a SQL query in the 'snowflakeassignment.sql' file:

```

68 FROM listing
69 LIMIT 10;
70
71 SELECT COUNT(*) AS listing_row_count
72 FROM listing;
73

```

Below the query editor, the 'Results (just now)' section shows a table with one row and one column:

LISTING_ROW_COUNT
10

The interface also includes a top navigation bar with tabs for 'worksheet-1', 'workbook.sql', and 'snowflakeassignment.sql'. The bottom status bar shows the time as 2:58 PM on 5/29/2026.

6. Submission checklist

- Employee table created in bootcamp.snow.
- Employee.csv records loaded into Snowflake Employee table.
- Employee data unloaded from Snowflake to s3://matrika-bucket/Snowflake/employee_data.csv.
- Redshift listing table unloaded to s3://matrika-bucket/Snowflake/listing_parquet/ in parquet format.
- Snowflake listing table created with individual columns.
- Parquet data loaded into Snowflake listing table using MATCH_BY_COLUMN_NAME = CASE_INSENSITIVE.
- Screenshots added for query results, S3 output, and verification.